

TMA4410 – Matematikk 2A

Uke 10

Frode Rønning

Stoff som er dekket i uke 10

- Lærebok: A. Quarteroni, F. Saleri, & P. Gervasio: Scientific Computing with MATLAB and Octave
 - Kap. 4: 4.2, 4.3
 - Kap. 8: 8.7, 8.8, 8.9

Runge-Kutta metoder generelt

- $u_{n+1} = u_n + h \sum_{i=1}^s b_i K_i$ der
 $K_i = f(t_n + c_i h, u_n + h \sum_{j=1}^s a_{ij} K_j), i = 1, 2, \dots, s$

c	A
	b

Butcher tablå

$$\mathbf{c} = [c_1, \dots, c_s]$$

$$\mathbf{b} = [b_1, \dots, b_s]$$

$$A = \begin{bmatrix} a_{11} & \dots & a_{1s} \\ \vdots & & \vdots \\ a_{s1} & \dots & a_{ss} \end{bmatrix}$$

Eksplisitt metode når $a_{ij} = 0$ for $j \geq i$ (på og over diagonalen)



John Butcher (1933 -)

Runge-Kutta av orden 2, 2 nivåer

- $u_{n+1} = u_n + h \sum_{i=1}^s b_i K_i$ der
 $K_i = f(t_n + c_i h, u_n + h \sum_{j=1}^s a_{ij} K_j), i = 1, 2, \dots, s$

- Trapesmetoden (Crank-Nicolsons metode):

- $u_{n+1} = u_n + \frac{h}{2} (K_1 + K_2)$ der
 $K_1 = f(t_n, u_n)$ og $K_2 = f\left(t_n + h, u_n + \frac{h}{2} (K_1 + K_2)\right)$
 $b_1 = b_2 = 1/2, c_1 = 0, c_2 = 1, A = \begin{bmatrix} 0 & 0 \\ 1/2 & 1/2 \end{bmatrix}$

- Forbedret Euler (Heuns metode)

- $u_{n+1} = u_n + \frac{h}{2} (K_1 + K_2)$ der
 $K_1 = f(t_n, u_n)$ og $K_2 = f(t_n + h, u_n + h K_1)$
- $b_1 = b_2 = 1/2, c_1 = 0, c_2 = 1, A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

0	0	0
1	1/2	1/2
	1/2	1/2

Butcher tablå

0	0	0
1	1	0
	1/2	1/2

Butcher tablå

Et ekstra blikk på Heuns metode

- $u_{n+1} = u_n + \frac{h}{2} \left(f(t_n, u_n) + f(t_n + h, u_n + hf(t_n, u_n)) \right)$

- Dette skjemaet kan vi se på slik (prediktor-korrektor-par):

1. $u_{n+1}^* = u_n + hf(t_n, u_n)$

Prediktor

2. $u_{n+1} = u_n + \frac{h}{2} \left(f(t_n, u_n) + f(t_n + h, u_{n+1}^*) \right)$

Korrektor

- Eks.: $y' = -2ty, y(0) = 1$

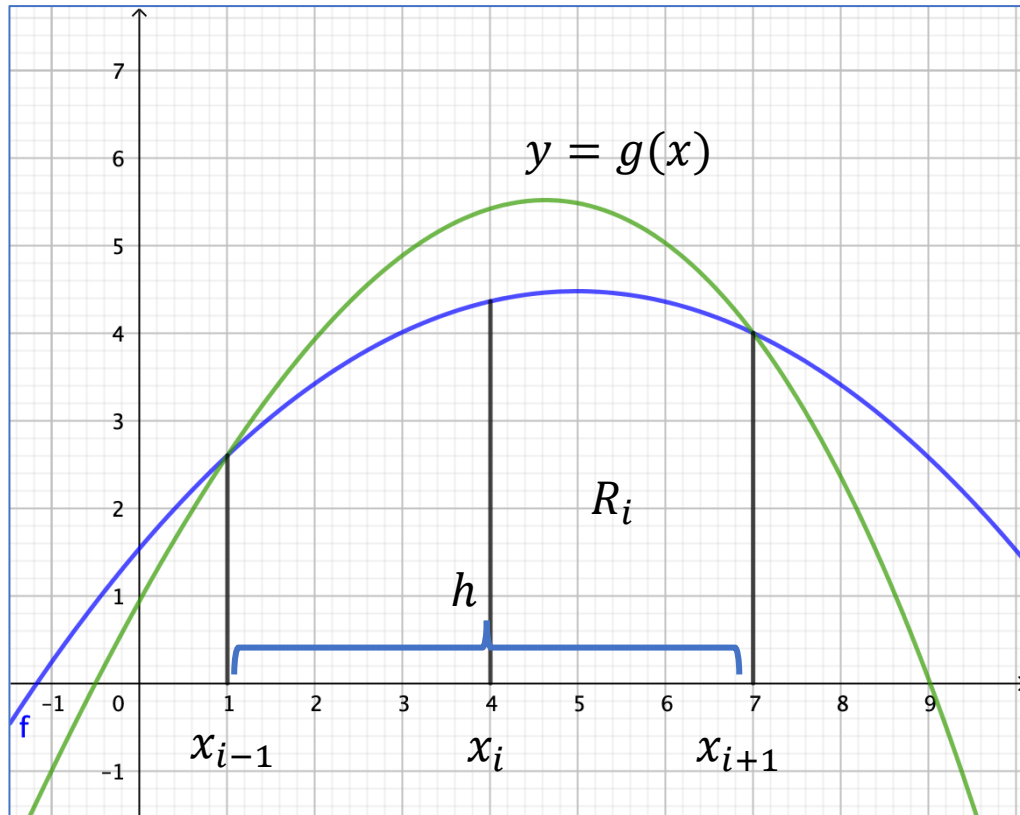
Integrere andregradspolynom

1. Sett $g(t) = t^2 + bt + c$ og regn ut $\int_0^x g(t)dt$
2. Regn ut $g(0) + 4g(x/2) + g(x)$

Hva observerer du?

Simpsons formel for integrasjon

- $\int_a^b g(x)dx \approx \frac{h}{6} (g(x_0) + 4g(x_1) + 2g(x_2) + \dots + 4g(x_{2n-1}) + g(x_{2n}))$



$$\left| \int_{x_{i-1}}^{x_{i+1}} g(x)dx - R_i \right| \leq \frac{K}{180} \frac{1}{16} h^4 \text{ der } K \geq |g^{(4)}(x)|$$

Runge-Kutta av orden 4, 4 nivåer

- $u_{n+1} = u_n + h \sum_{i=1}^s b_i K_i$ der
 $K_i = f(t_n + c_i h, u_n + h \sum_{j=1}^s a_{ij} K_j), i = 1, 2, \dots, s$

- RK4

- $u_{n+1} = u_n + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$ der
 $K_1 = f(t_n, u_n), K_2 = f\left(t_n + \frac{h}{2}, u_n + \frac{h}{2} K_1\right),$
 $K_3 = f\left(t_n + \frac{h}{2}, u_n + \frac{h}{2} K_2\right)$ og $K_4 = f(t_n + h, u_n + h K_3)$

- $b_1 = 1/6, b_2 = 1/3, b_3 = 1/3, b_4 = 1/6,$

$$c_1 = 0, c_2 = 1/2, c_3 = 1/2, c_4 = 1, A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

0	0	0	0	0
1/2	1/2	0	0	0
1/2	0	1/2	0	0
1	0	0	1	0
	1/6	1/3	1/3	1/6

Butcher tablå

Orden og antall nivåer av eksplisitte RK-metoder

- La $p(s)$ være maksimal orden som kan oppnås for en eksplisitt RK-metode med s nivåer. Da gjelder

$$p(1) = 1 \text{ (Forlengs Euler)}$$

$$p(2) = 2 \text{ (Heuns metode)}$$

$$p(3) = 3$$

$$p(4) = 4 \text{ (RK4)}$$

$$p(5) = 4$$

$$p(6) = 5$$

$$p(7) = 6$$

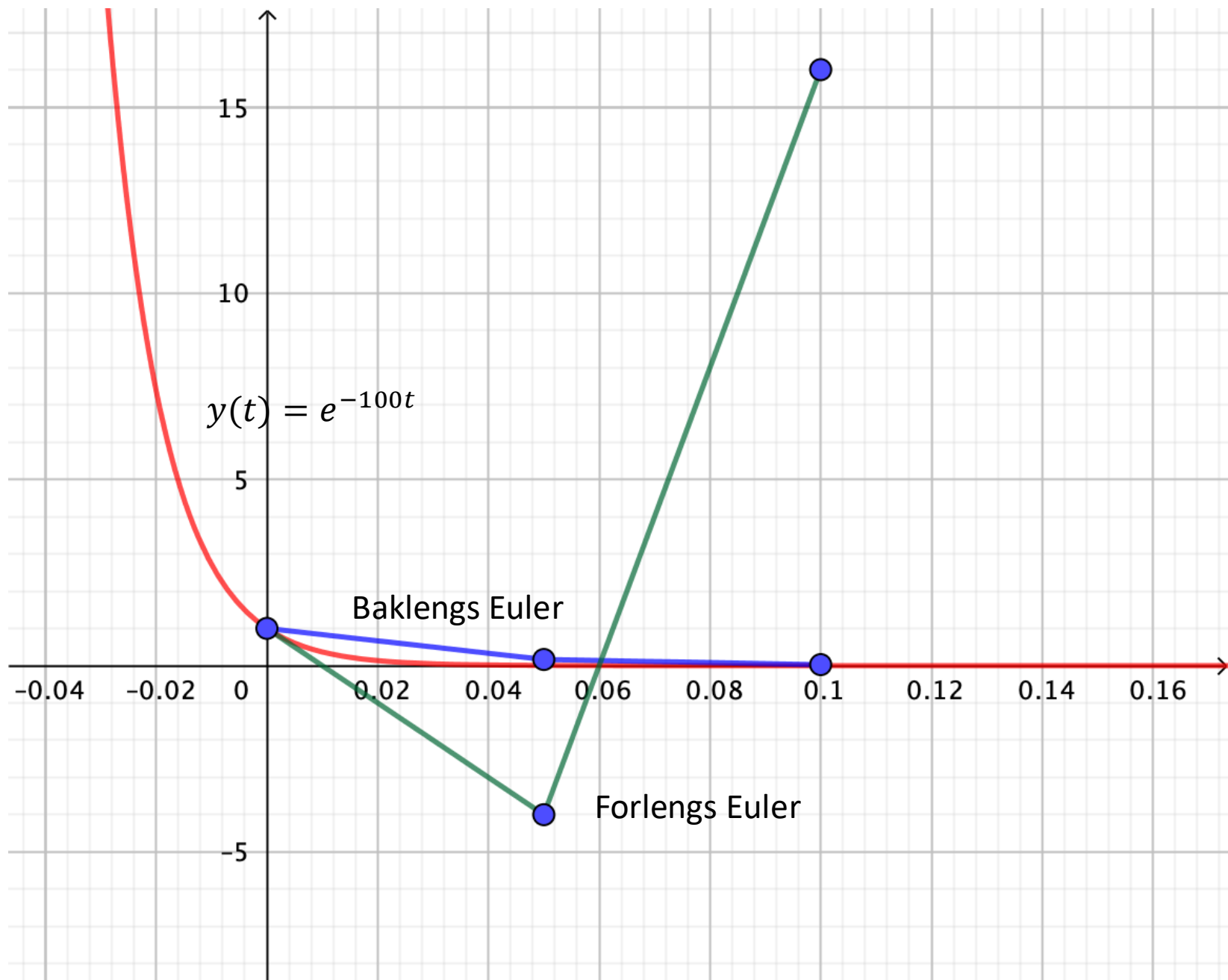
$$p(8) = 6$$

$$p(s) \leq s - 2, s \geq 9$$

(Butcher, 1964, 1965)

Litt om stabilitet

- eller, hvorfor skal vi bry oss med implisitte metoder?
- Forlengs Euler:
 - $u_{n+1} = u_n + hf(t_n, u_n)$
- Baklengs Euler:
 - $u_{n+1} = u_n + hf(t_n + h, u_{n+1})$
- Eks.: $y' = -100y, y(0) = 1$ har eksakt løsning $y(t) = e^{-100t}$
- Med $h = 0.05$ gjør to skritt med hver av disse metodene



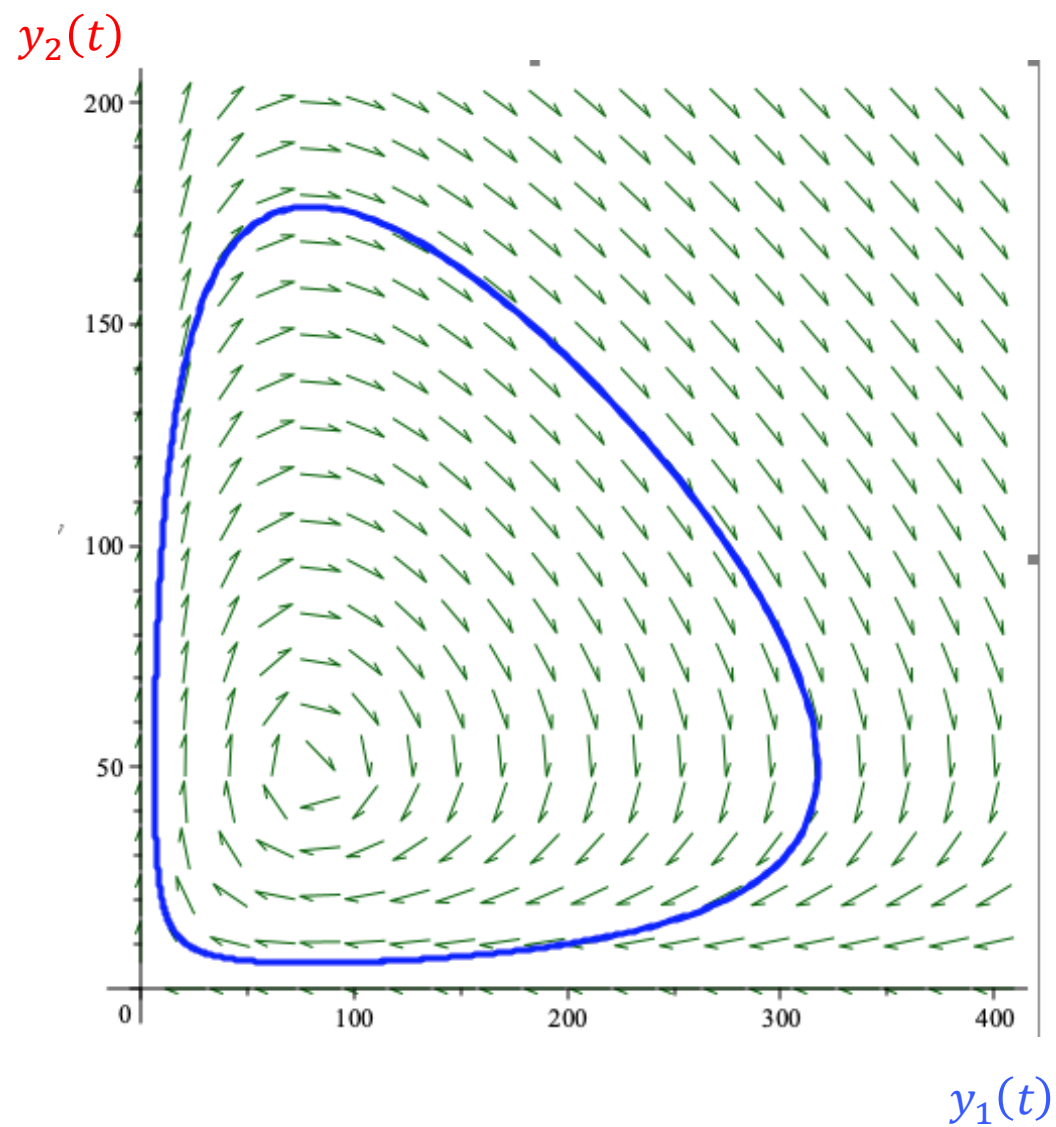
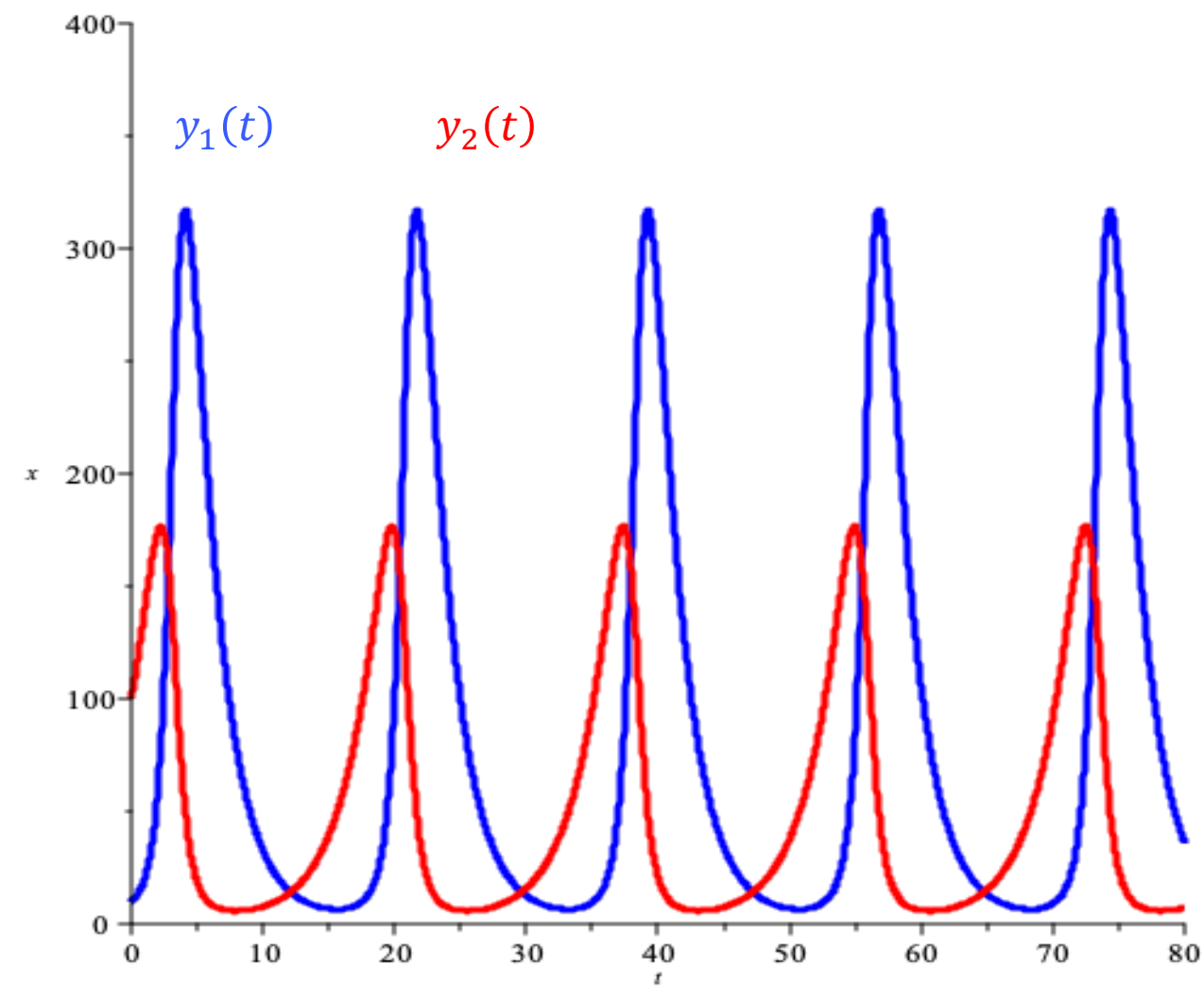
System av ikke-lineære differensialligninger

- Eksempel (Lotka-Volterras ligninger):

$$y_1'(t) = -0.5y_1(t) + 0.01y_1(t)y_2(t), \quad y_1(0) = 10$$

$$y_2'(t) = 0.4y_2(t) - 0.005y_1(t)y_2(t), \quad y_2(0) = 100$$

- Gjør ett skritt med forlengs Euler, $h = 0.1$
- Finn likevektsløsningene ($y_1'(t) = y_2'(t) = 0$)



Numerisk derivasjon

1. $\frac{g(x+h)-g(x-h)}{2h} = g'(x) + \frac{h^2}{12} (g'''(c_1) + g'''(c_2)) = g'(x) + \mathcal{O}(h^2)$
2. $\frac{g(x+h)-2g(x)+g(x-h)}{h^2} = g''(x) + \frac{h^2}{24} (g^{(4)}(c_1) + g^{(4)}(c_2)) = g''(x) + \mathcal{O}(h^2)$

Bevis: Taylors teorem

Å løse andreordens ligninger

- Gitt $y'' = f(t, y(t), y'(t))$, $y(t_0) = \alpha_0$, $y'(t_0) = \beta_0$
- To muligheter:
 1. Sett $y'(t) = z(t)$ og løs systemet
$$\begin{aligned} y' &= z(t) \\ z' &= f(t, y, z) \end{aligned}; y(t_0) = \alpha_0, z(t_0) = \beta_0$$
 2. Leapfrogmetoden

Leapfrogmetoden

- Gitt $y'' = f(t, y(t), y'(t))$, $y(t_0) = \alpha_0$, $y'(t_0) = \beta_0$
- Inspirert av approksimasjonsformlene for 1. og 2. deriverte, sett

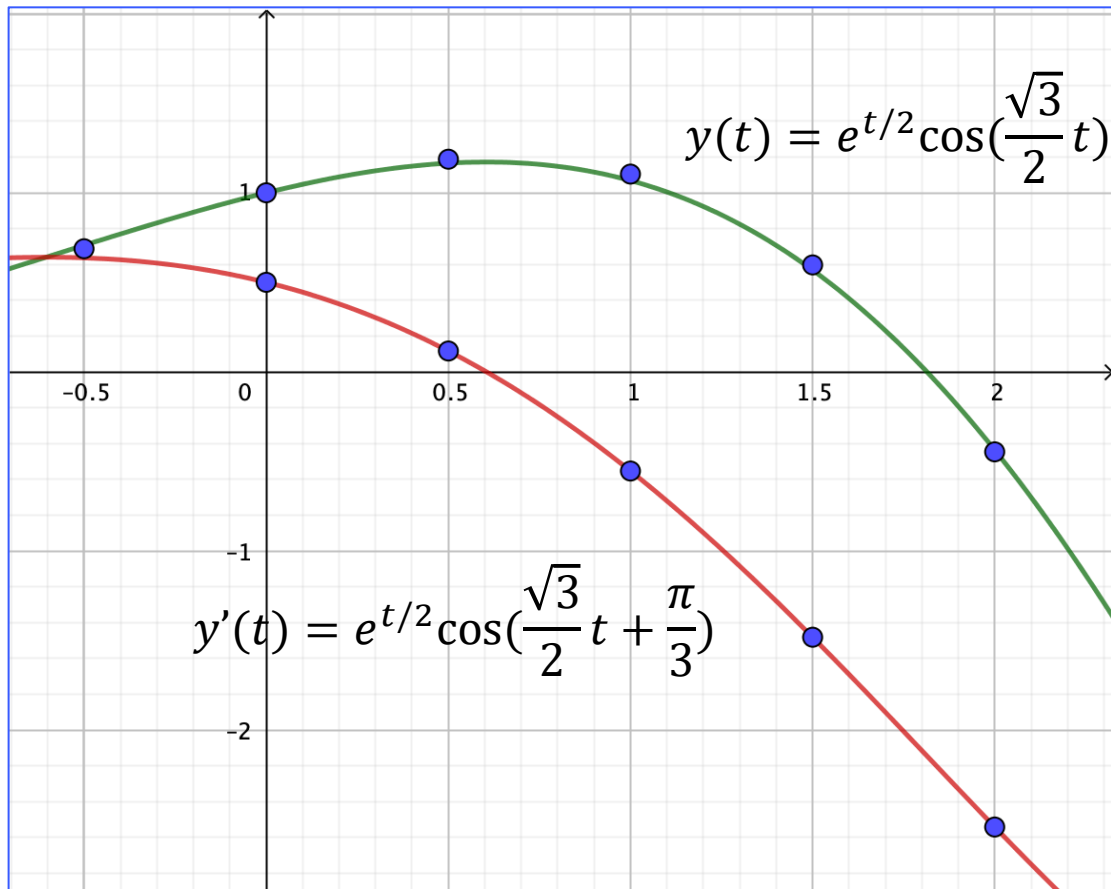
$$1. \quad \frac{u_{n+1} - 2u_n + u_{n-1}}{h^2} = f(t_n, u_n, v_n)$$

$$2. \quad v_n = \frac{u_{n+1} - u_{n-1}}{2h}$$

- $u_0 = \alpha_0, v_0 = \beta_0$
- Flerskrittmetode: Trenger to startverdier for å komme i gang

Eksempel

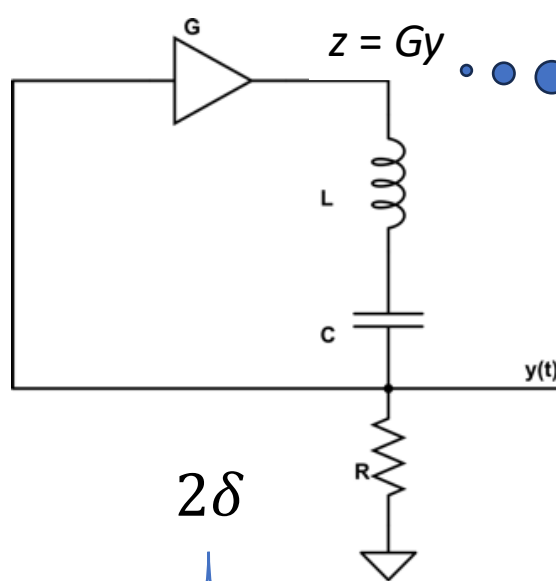
- Gitt $y''(t) - y'(t) + y(t) = 0, y(0) = 1, y'(0) = \frac{1}{2}$
- Eksakt løsning: $y(t) = e^{t/2} \cos(\frac{\sqrt{3}}{2} t), y'(t) = e^{t/2} \cos(\frac{\sqrt{3}}{2} t + \frac{\pi}{3})$
- Leapfrog:
 1. $\frac{u_{n+1} - 2u_n + u_{n-1}}{h^2} = v_n - u_n$
 2. $v_n = \frac{u_{n+1} - u_{n-1}}{2h}$
 - $u_0 = 1, v_0 = \frac{1}{2}$. Sett $h = 1/2$ og finn tilnærmingsverdier til $y(2)$ og $y'(2)$
 - For å komme i gang må vi finne en verdi for u_{-1}
- $u_{-1} = \frac{1}{2} [(h^2 - 2h)v_0 + (2 - h^2)u_0] = \frac{11}{16} \approx 0.6875$
- Dette gir $u_{n+1}(1 - h/2) = (2 - h^2)u_n - (1 + h/2)u_{n-1}, u_{-1} = \frac{11}{16}, u_0 = 1$
$$v_n = \frac{u_{n+1} - u_{n-1}}{2h}$$



i	t_i	u_i	v_i	y(t_i)	y'(t_i)
-1	-0,5	0,6875			
0	0	1,0000	0,5000	1,0000	0,5000
1	0,5	1,1875	0,1042	1,1655	0,1162
2	1	1,1042	-0,5903	1,0681	-0,5536
3	1,5	0,5972	-1,5509	0,5683	-1,4820
4	2	-0,4468	-2,6350	-0,4364	-2,5418
5	2,5	-2,0378		-1,9542	-3,4816

Leapfrog Eksakt

Oscillatorkrets med lineær forsterker



$$z = Gy$$

Lineær
funksjon

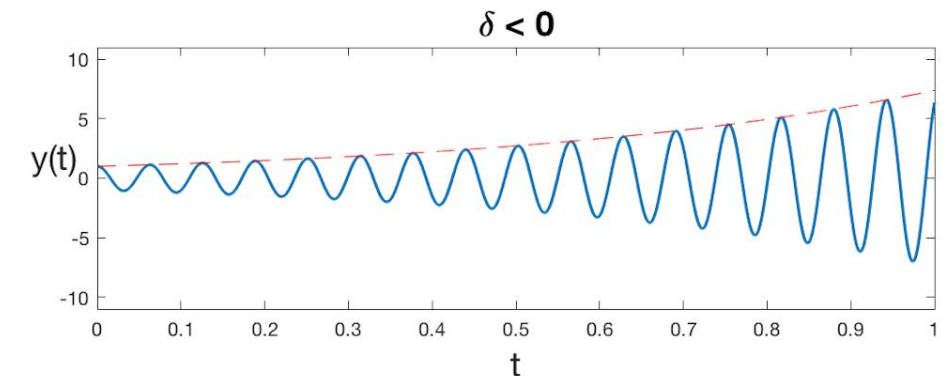
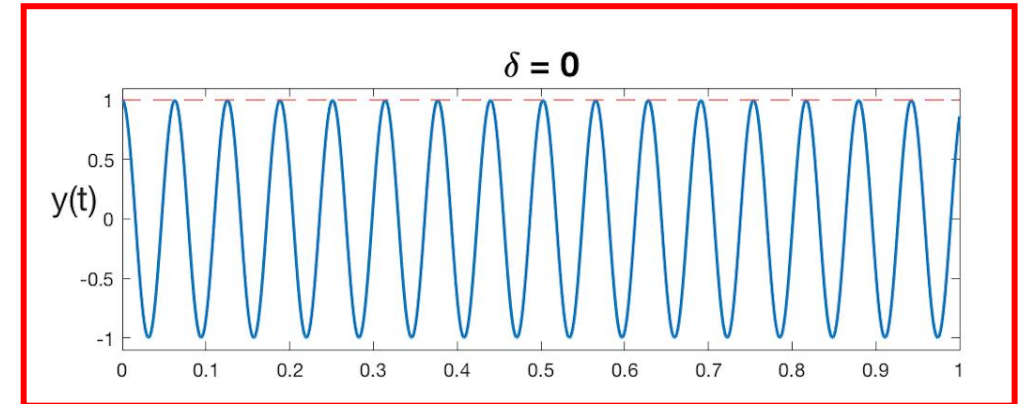
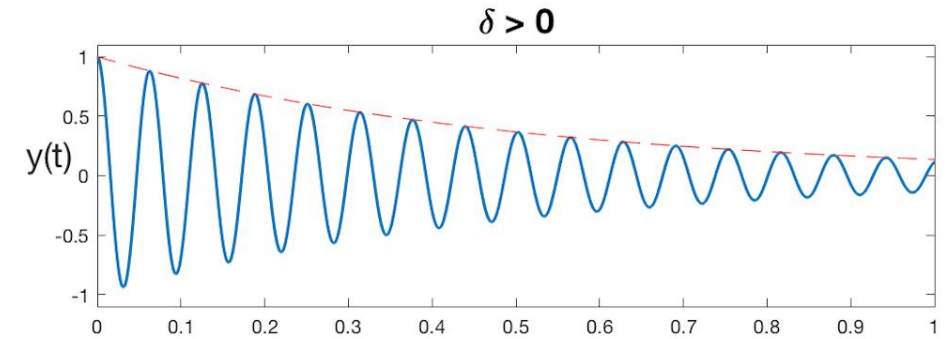
Ønsker dette

$$y'' + (1 - G)\frac{R}{L}y' + \frac{1}{LC}y = 0$$

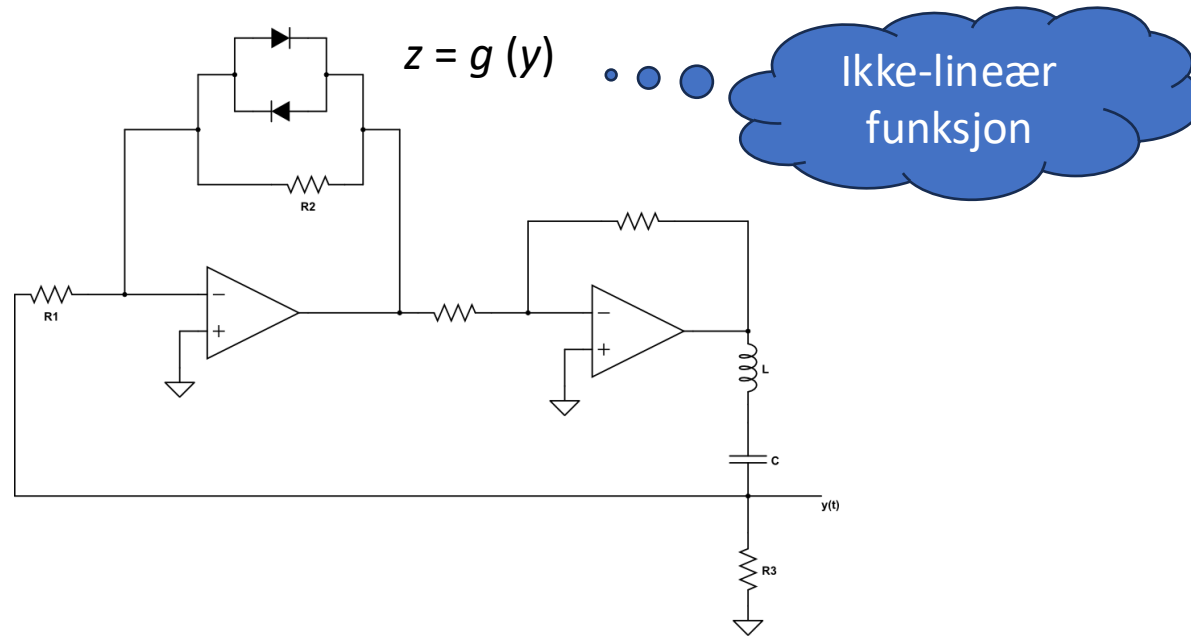
$$y' = z$$

$$z' = (G - 1)\frac{R}{L}z + \frac{1}{LC}y$$

Lineært
system



Oscillatorkrets med ikke-lineær forsterker



$$y'' + \frac{R_3}{L} f(y) y' + \frac{1}{LC} y = 0$$

$$\text{der } f(y) = 1 - \frac{dg(y)}{dy}$$

Ikke-lineært system

$$y' = z$$

$$z' = -\frac{1}{LC} y - f(y) \frac{R_3}{L} z$$

